



Science

Stage 8

Paper 1

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Draw a straight line to match each **blood component** to its correct **function**.

Draw three lines only.

blood component

function

red blood cell

carries carbon dioxide

white blood cell

fights pathogens

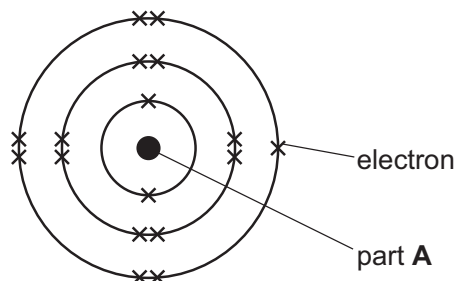
plasma

digests food

transports oxygen

[2]

- 2 Look at the model of a diagram of an atom.



- (a) Write down the name of part **A**.

..... [1]

- (b) Which **two** particles make up part **A**?

..... and [1]

- (c) Which scientist developed the nuclear or planetary model of the atom?

..... [1]

- (d) Write down the charge on an electron.

..... [1]

- (e) The electrons in an atom are held in position.

They do **not** escape from the atom.

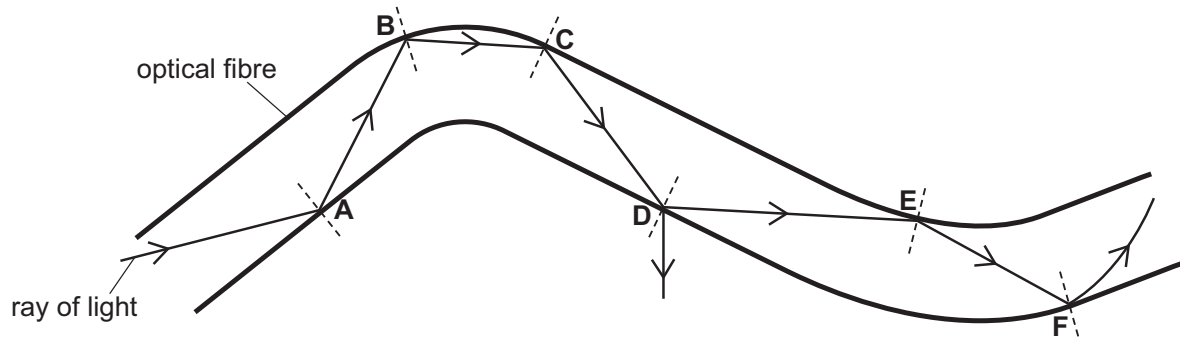
Explain how electrons are held in position.

Use ideas about the charges on the particles.

.....

 [2]

- 3 Mia draws the path of a ray of light through an optical fibre.



A ray of light travels through an optical fibre without any of the ray escaping outside of the fibre.

Mia makes **three** mistakes in her drawing.

Complete the sentences to describe her mistakes at **B**, **D** and **F**.

The mistake at **B**

.....

The mistake at **D**

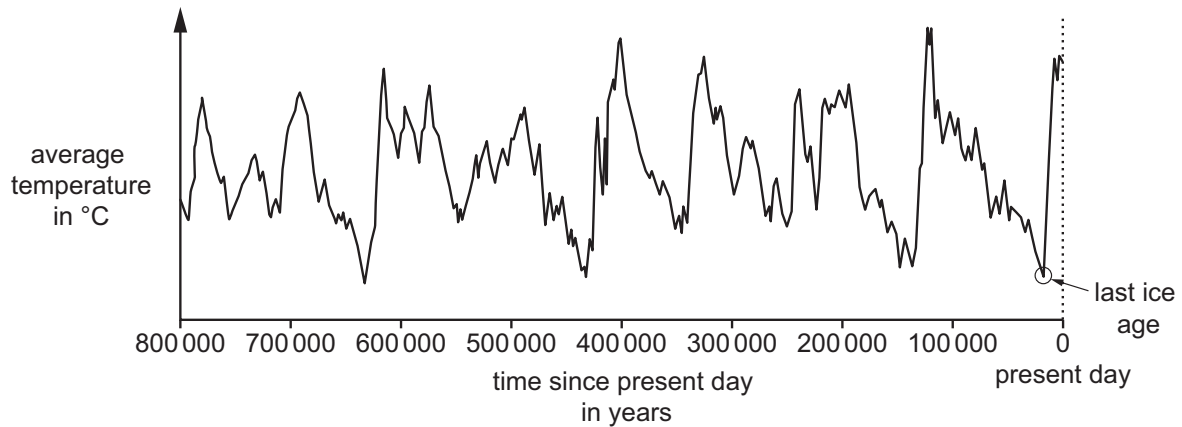
.....

The mistake at **F**

.....

[3]

- 4 Look at the graph showing how the average temperature of the surface of the Earth has changed over the last 800 000 years.



- (a) The last ice age is circled on the graph.

Circle a point on the graph when **one other** ice age happened.

[1]

- (b) Describe the pattern shown in the average temperature of the surface of the Earth over the past 800 000 years.

.....

.....

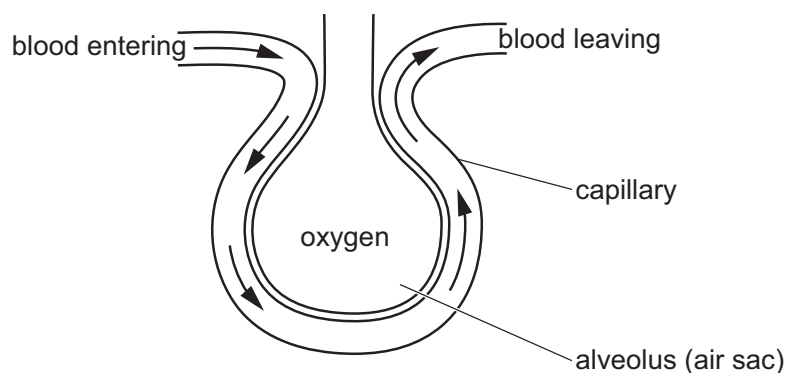
..... [1]

- (c) Changes in the percentage of some gases in the atmosphere cause climate change.

Name **one** gas that causes climate change.

..... [1]

- 5 (a) Look at the diagram showing a blood capillary carrying blood to and from an alveolus (air sac) in the lungs.



- (i) Oxygen in the alveolus (air sac) enters the blood.

Write down the name of this process.

..... [1]

- (ii) Blood contains dissolved carbon dioxide.

Describe what happens to the carbon dioxide in the blood capillary.

.....
 [1]

- (b) The table shows the percentage of gases in air as it is breathed in and breathed out.

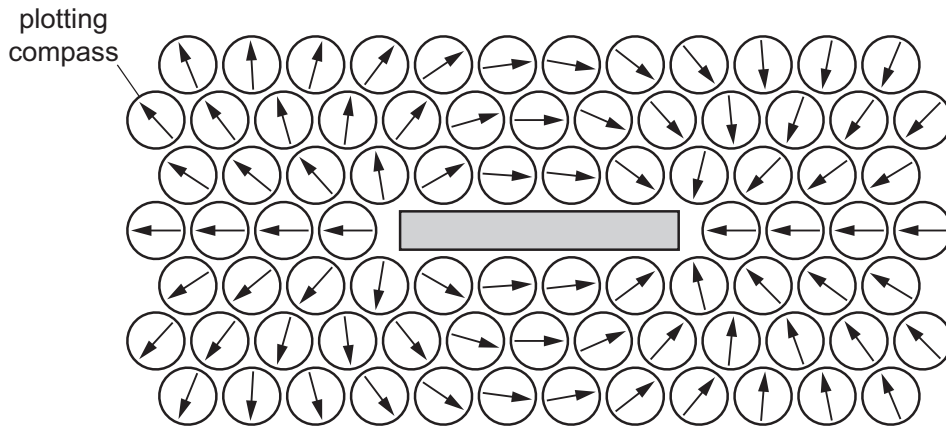
gas	breathed in %	breathed out %
carbon dioxide	0.04	
other gases	78.96	78.96
oxygen	21.00	

Predict the percentages of carbon dioxide and oxygen in **breathed out** air.

Write your answers in the table.

[2]

- 6 Carlos puts plotting compasses around a magnet.



- (a) Look at the plotting compasses.

Describe what the plotting compasses show.

.....

.....

..... [2]

- (b) The magnet is in the middle of the plotting compasses.

Which diagram shows the position of the poles of the magnet?

Circle the correct answer.

N S

N N

S N

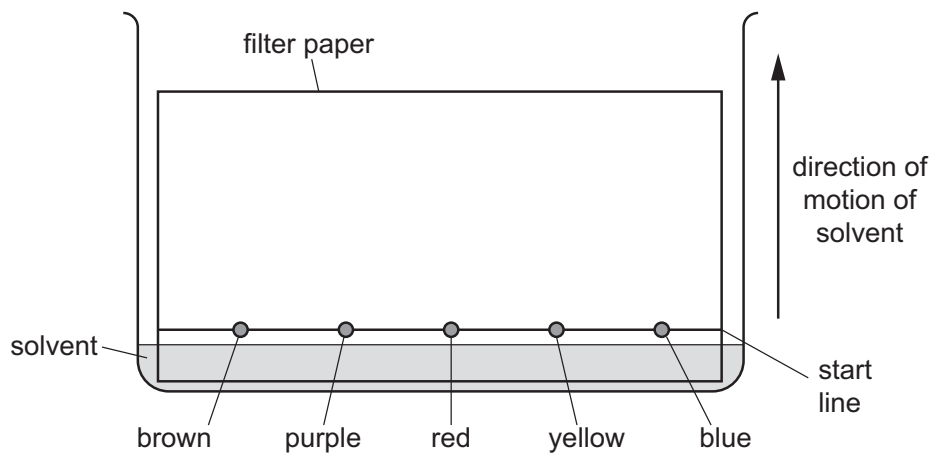
S S

[1]

7 Oliver separates some food colourings.

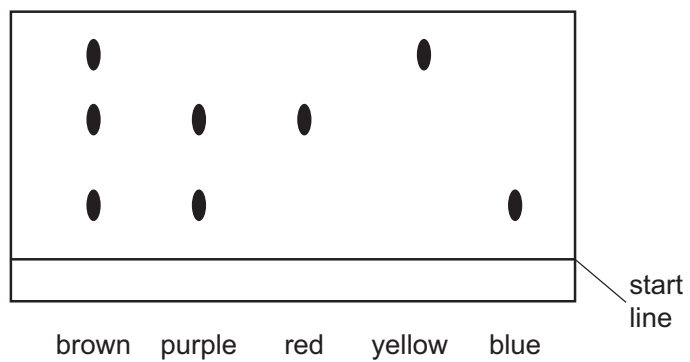
Oliver:

- draws a start line in pencil on some filter paper
- places dots of different food colourings on the start line
- puts the filter paper in the solvent as shown in the diagram



- waits until the solvent has soaked up the filter paper.

Look at his results.



(a) Write down the name of this separation technique.

..... [1]

(b) Which food colourings are **not** pure substances?

.....

Explain your answer.

.....

.....

[2]

(c) Write down the colours in the purple food colouring.

Choose from the list.

brown

red

yellow

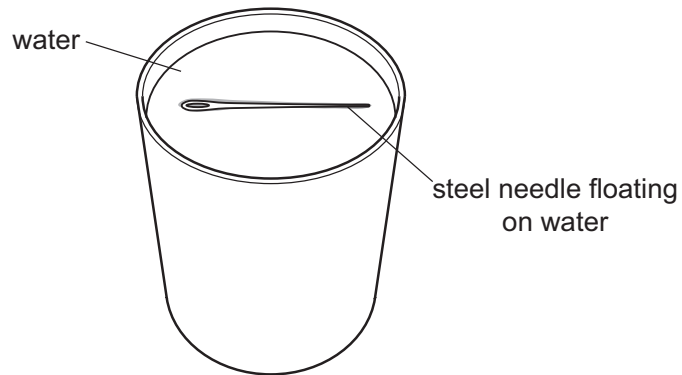
blue

..... [1]

8 Angelique investigates a steel needle floating on water.

In her first experiment Angelique:

- passes a steel needle over a strong magnet
- floats the steel needle on the surface of the water.



She repeats the experiment five times.

Each time, she floats the steel needle in different directions on the surface of the water.

The needle turns to point in the same direction every time.

Explain why.

.....

.....

.....

..... [2]

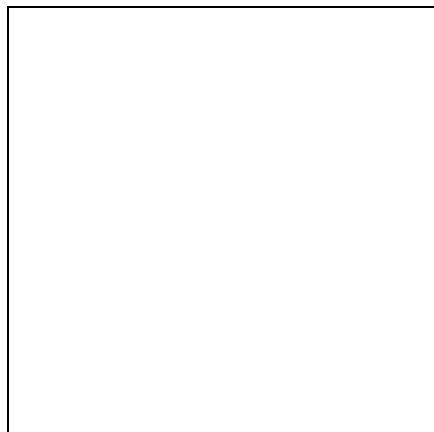
- 9 Pure substances contain either one element **or** one compound.

Impure substances are mixtures.

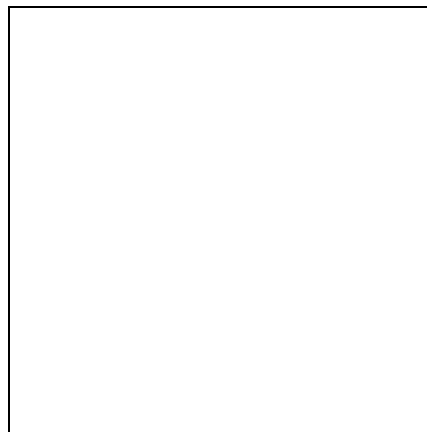
- (a) Draw diagrams to show a model of the molecules in pure water **and** in impure water.

A molecule of water is represented by ○

A molecule of sugar is represented by ●



pure water



impure water

[2]

- (b) Which word describes using a model?

Circle the correct answer.

analogy

conclusion

precision

symbol

[1]

10 Safia plays a game of tennis.

(a) Look at the table showing the percentages of three nutrients in four foods.

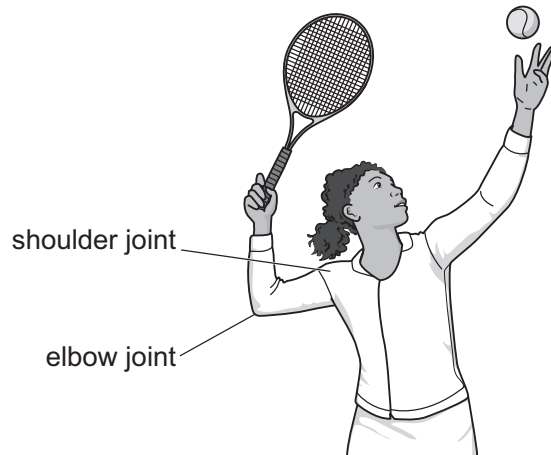
food	protein %	carbohydrate %	fat %
banana	4	88	2
boiled egg	13	1	11
cola drink	0	10	0
chicken sandwich	19	24	6

To give her energy, Safia eats 100 g of one of the foods before she plays her game of tennis.

Which food in the table contains the most energy?

..... [1]

(b) When Safia hits the ball, her shoulder joint and her elbow joint move.



(i) Write down the name of the type of joint at the shoulder.

..... [1]

(ii) Explain how antagonistic muscles allow movement around the elbow joint.

.....
 [1]

(c) Three of Safia's friends make predictions about her breathing rate.

I predict her breathing rate at the start will be **lower** than at the end of the game.



Lily

I predict her breathing rate at the start will be **higher** than at the end of the game.



Gabriella

I predict her breathing rate at the start will be the **same** as at the end of the game.



Mike

Circle which friend has the correct prediction.

Lily

Gabriella

Mike

Explain your answer.

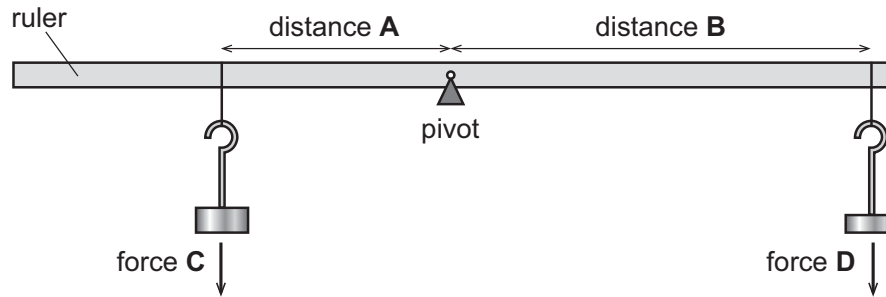
.....

.....

[2]

11 Chen investigates turning forces.

He balances a ruler using different forces and distances.



(a) The turning force is called a moment.

Describe how to calculate a moment.

.....

..... [1]

(b) Chen makes the ruler balance three times.

He writes down the different measurements each time.

distance A	= 6 cm,	distance B	= 12 cm,	force C	= 4 N,	force D	= 2 N
distance A	= 5 cm,	distance B	= 2 cm,	force C	= 4 N,	force D	= 10 N
distance A	= 4 cm,	distance B	= 8 cm,	force C	= 2 N,	force D	= 1 N

(i) Complete the table using these results.

distance A in cm	force C in N	distance B in cm	

[3]

(ii) Chen says,

‘I think my results are reliable.’

Tick (✓) to show if Chen is correct.

Yes ☐ No ☐

Explain your answer.

.....

.....

[1]

12 This question is about energy resources.

(a) Look at the list of resources.

bioplastics

fossil fuels

tidal power

vegetable oil

solar power

Complete the table to group the resources into renewable **and** non-renewable resources.

renewable	non-renewable

[2]

(b) Wind power and coal are resources used to generate electricity.

Describe **two** advantages of using wind power instead of coal to generate electricity.

1

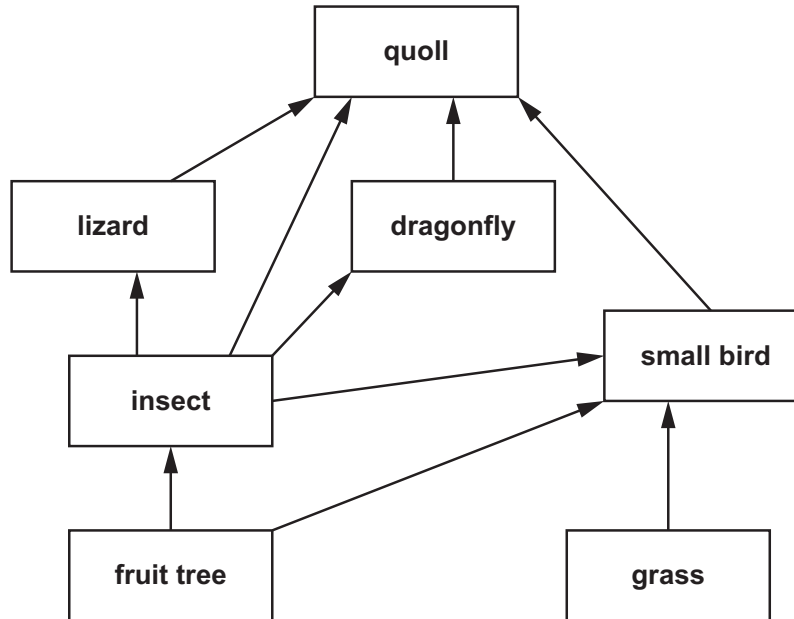
.....

2

.....

[2]

13 Look at the food web for an Australian grassland ecosystem.



(a) Cats are an invasive species in Australia.

Cats eat small birds and lizards.

Explain the effect the introduction of cats has on the dragonfly population in this food web.

.....

.....

.....

..... [2]

(b) Insects eat fruit.

Farmers use a toxic substance to kill insects that eat the fruit on their fruit trees.

Quolls are also killed by this toxic substance.

Explain why.

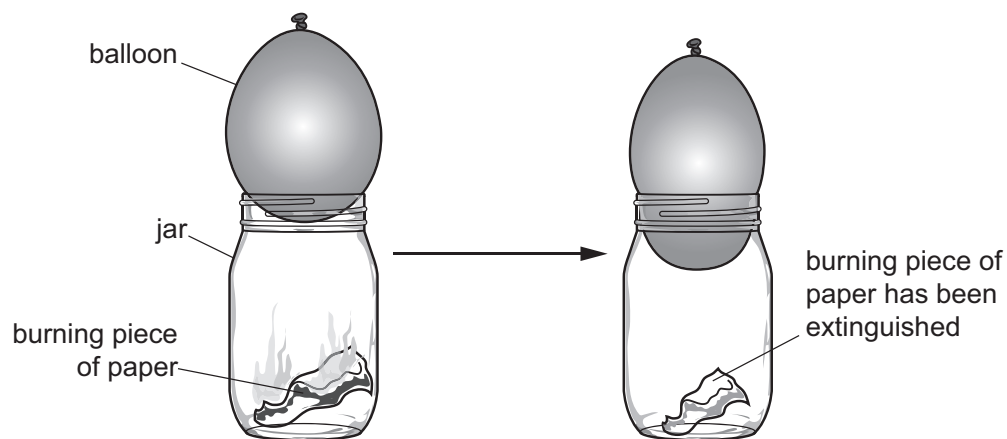
.....

..... [1]

14 A teacher investigates pressure.

The teacher:

- fills a balloon with water
- places a burning piece of paper in a jar
- puts the balloon on top of the jar.



(a) Describe how the teacher makes this investigation safe.

.....
 [1]

(b) When the piece of paper stops burning, the temperature in the jar decreases.

The air pressure in the jar decreases.

Use the particle theory to explain why.

.....

 [2]

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The Periodic Table of Elements

Group												
1	2						3	4	5	6	7	8



Science

Stage 8

Paper 2

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
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INFORMATION

- The total mark for this paper is 50.
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- 1 The table shows information about four different adults who play sports.

The adults are the same age and the same height.

adults	physical activity level	do they smoke?
W	low	no
X	high	no
Y	high	yes
Z	low	yes

- (a) (i) Which adult, **W**, **X**, **Y** or **Z**, is most likely to have health problems?

.....

[1]

- (ii) The four adults test their fitness by exercising.

After exercising, they measure their pulse rate.

What do they do to make the test fair?

Tick (✓) the correct answer.

two adults run and two adults walk for 5 minutes ☐

use a ruler to measure distance ☐

all four adults run at the same speed for 1 minute ☐

repeat the exercise four times ☐

[1]

- (b) (i) Adult **Y** takes a vitamin A tablet every morning.

What is the function of vitamin A?

..... [1]

- (ii) Adult **Z** eats a diet that has a very high fat content.

Write down **one** health problem caused by this type of diet.

..... [1]

- 2 Aiko reacts iron with dilute sulfuric acid.

She takes the temperature of the dilute sulfuric acid before she adds the iron.

She takes the temperature of the reaction mixture at the end of the reaction.

Look at her results.

temperature of dilute sulfuric acid at start = 18°C

temperature of reaction mixture at end = 28°C

- (a) Write down the **type** of reaction between iron and dilute sulfuric acid.

Choose from the list.

endothermic

exothermic

neutralisation

precipitation

type of reaction

Explain your answer.

.....

.....

[2]

- (b) Iron reacts with sulfuric acid.

Iron sulfate and hydrogen are made.

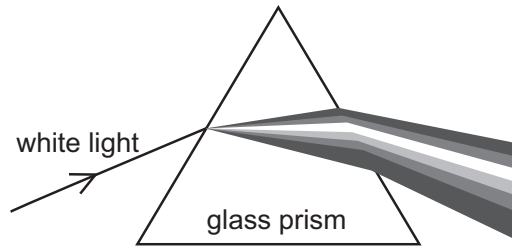
Write down the **word** equation for this reaction.

.....

[1]

3 White light is made of many colours.

(a) Oliver uses a glass prism and a ray of white light.



Complete the sentences.

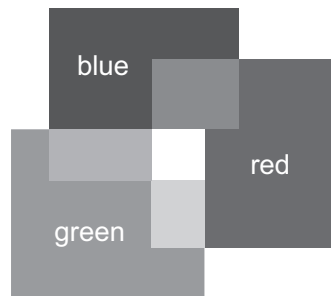
A ray of white light enters the glass prism.

This ray of white light changes speed because the light is

White light is split into different colours by the process of

[2]

(b) Oliver adds different colours of light together.



Complete the additions of light.

One has been done for you.

red + blue = magenta

red + green =

green + blue =

red + blue + green =

[2]

4 Name **two** objects that are found in all known galaxies.

1

2

[2]

5 The picture shows a woodland ecosystem.



(a) One habitat shown in the woodland ecosystem is tree roots.

Name **two other** habitats shown in the picture.

1

2

[2]

(b) Jamila observes a new species of bird in the woodland.

She makes a hypothesis,

‘The new species of bird reduces the number of other species of bird in the woodland.’

Jamila counts the number of species of bird in the woodland.

Six months later, she counts the number of species of bird in the same place in the woodland.

- (i) Write down **one other** factor she must keep the same during both counts.

..... [1]

- (ii) What is the dependent variable?

..... [1]

- (iii) Look at Jamila's table of results.

count number	number of species of bird
1	14
2	12

Do these results support her hypothesis?

.....

Give **two** reasons for your answer.

1

.....

2

.....

[2]

- 6 Look at the table showing the reactions of some metals with cold water and with dilute hydrochloric acid.

metal	reaction with cold water	reaction with dilute hydrochloric acid
A	no reaction	no reaction
B	floats on water, reacts quickly and gives off a gas	floats on acid, reacts very quickly and gives off a gas
C	reacts slowly and gives off a gas	reacts quickly and gives off a gas
D	no reaction	reacts slowly and gives off a gas

Look at this list of metals.

calcium

gold

magnesium

sodium

zinc

- (a) Identify the metals **A**, **B**, **C** and **D**.

Choose from the list.

Metal **A** is

Metal **B** is

Metal **C** is

Metal **D** is

[3]

- (b) Write down the name of the gas made when metals **B** and **C** react with water.

.....

[1]

- 7 Lily uses this equipment to make an electromagnet.



battery



copper wire



iron nail

- (a) Describe how Lily makes an electromagnet.

.....

.....

..... [2]

- (b) Which of these changes **increase** the strength of her electromagnet?

Tick (✓) the **two** correct answers.

cover the copper wire in plastic

☐

use a non metal for the wire

☐

use two batteries

☐

remove the iron nail

☐

increase the current

☐

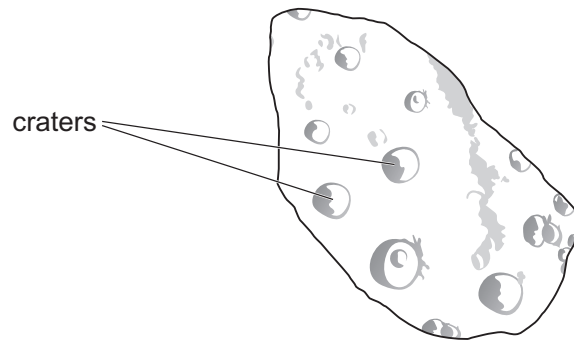
connect the equipment for a shorter time

☐

[2]

- 8 The picture shows an asteroid.

Asteroids are found in space.



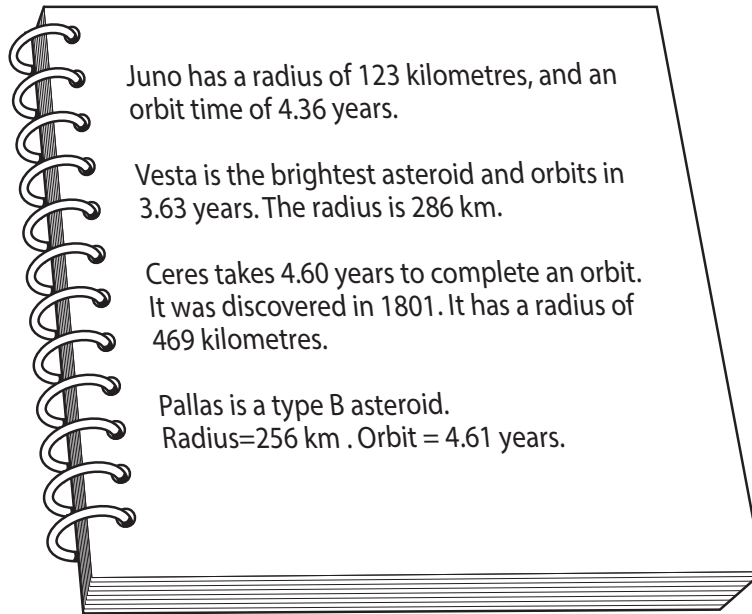
- (a) (i) What is an asteroid?

.....
..... [1]

- (ii) Suggest how craters are made on asteroids.

.....
..... [1]

(b) Pierre writes down some information about four asteroids.



He draws a table to show the information more clearly.

(i) Present his information on orbit time and radius for each asteroid in the table.

[3]

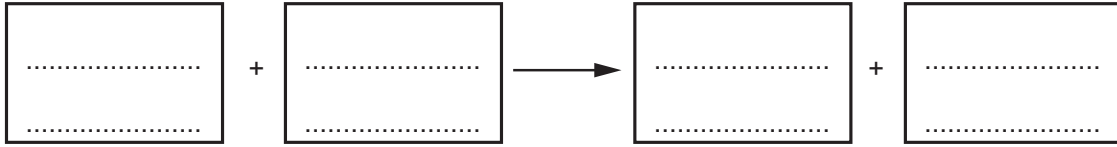
(ii) All four asteroids travel the same distance in one year.

Which asteroid travels at the **greatest** mean speed?

..... [1]

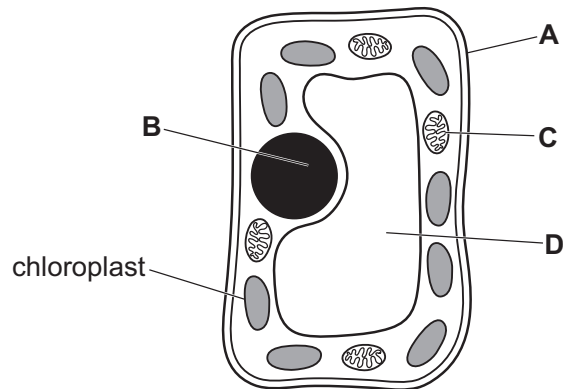
9 This question is about aerobic respiration.

(a) Write down the **word** equation for aerobic respiration.



[2]

(b) Carlos draws a diagram of a plant cell.



Where does aerobic respiration happen in this cell?

Choose from the letters **A**, **B**, **C** or **D**.

letter

[1]

10 Blessy dissolves salt in water to make a salt solution.

She makes four solutions.

Look at the quantities she uses.

solution	mass of salt in g	volume of water in cm ³
A	1	10
B	6	30
C	5	20
D	10	50

(a) Which solution is the **most** concentrated?

Circle the correct answer.

A

B

C

D

Explain your answer.

.....

[2]

(b) Describe how the solubility of salt changes as the temperature increases.

..... [1]

11 Ahmed has time and distance measurements for a bicycle journey.

time in s	distance in m
0	0
20	60
40	240
60	420
80	600
100	760

(a) Write down the equipment he uses to make these measurements.

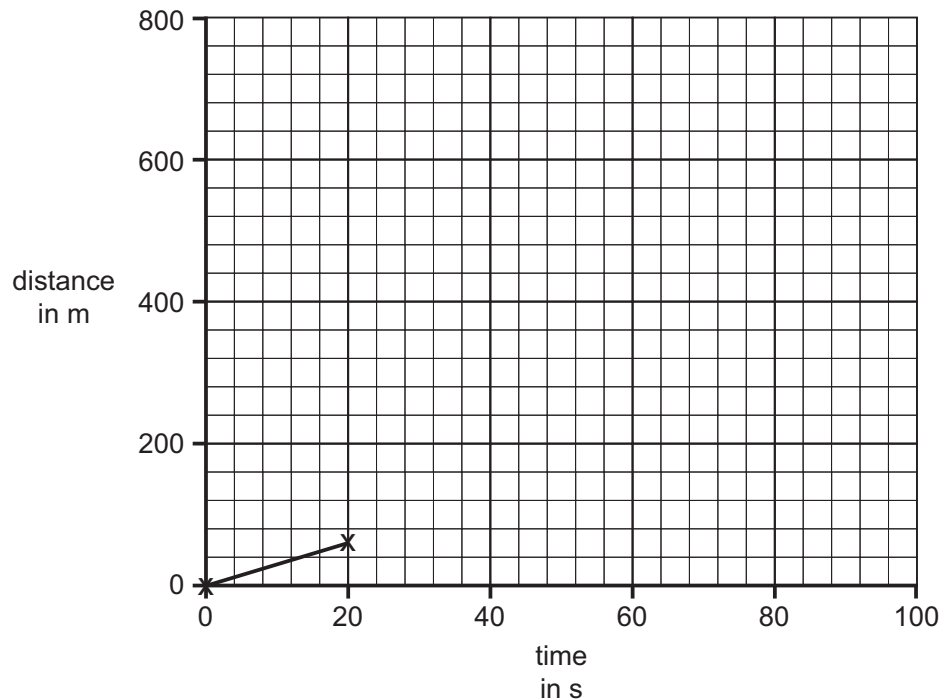
time to measure 20 s

distance to measure 60 m

[2]

(b) Complete the graph by:

- plotting the other four points
- drawing a line to connect the points.



[2]

(c) Calculate the speed when a distance of 60 m is travelled in 20 s.

Use the equation shown.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{speed} = \dots\dots\dots \text{ m/s}$$

[1]

12 Here are some sentences about climate and weather.

Tick (✓) each correct sentence.

Climate is the day to day changes in the atmosphere.

☐

Climate changes over a long period of time.

☐

Weather changes over a short period of time.

☐

Weather gives information about the atmosphere.

☐

[2]

13 Priya uses chemical formulae to model chemical compounds.

The chemical formula of sodium carbonate is Na_2CO_3 .

The formula of sodium carbonate contains:

- 2 atoms of sodium, Na
- 1 atom of carbon, C
- 3 atoms of oxygen, O.

(a) The formula of carbon dioxide contains:

- 1 atom of carbon, C
- 2 atoms of oxygen, O.

Write down the chemical formula for carbon dioxide.

formula

[1]

(b) The formula of nitric acid contains:

- 1 atom of hydrogen, H
- 1 atom of nitrogen, N
- 3 atoms of oxygen, O.

Write down the chemical formula of nitric acid.

formula

[1]

(c) Look at the word equation.



The formula of sodium nitrate is NaNO_3 .

The formula of water is H_2O .

Write down the **symbol** equation for this reaction.

..... [2]

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Group																			
1	2	Key						3	4	5	6	7	8						
		atomic number atomic symbol name relative atomic mass						1 H hydrogen 1											
								2 He helium 4											
3	Li lithium 7	4	Be beryllium 9					5	B boron 11	6	C carbon 12	7	N nitrogen 14	8	O oxygen 16	9	F fluorine 19	10	Ne neon 20
11	Na sodium 23	12	Mg magnesium 24					13	Al aluminium 27	14	Si silicon 28	15	P phosphorus 31	16	S sulfur 32	17	Cl chlorine 35.5	18	Ar argon 40
19	K potassium 39	20	Ca calcium 40					31	Ga gallium 70	32	Ge germanium 73	33	As arsenic 75	34	Se selenium 79	35	Br bromine 80	36	Kr krypton 84
37	Rb rubidium 85	38	Sr strontium 88					49	In indium 115	50	Sn tin 119	51	Sb antimony 122	52	Te tellurium 128	53	I iodine 127	54	Xe xenon 131
55	Cs caesium 133	56	Ba barium 137					81	Tl thallium 204	82	Pb lead 207	83	Bi bismuth 209	84	Po polonium —	85	At astatine —	86	Rn radon —
87	Fr francium —	88	Ra radium —					113	Nh nihonium —	114	Fl flerovium —	115	Mc moscovium —	116	Lv livermorium —	117	Ts tennessine —	118	Og oganesson —